NPRB2301 Sample and Data Update

June 9, 2025

Old Sharks

The eyes have it

This report summarizes the samples available for NPRB 2301, their status and the status of laboratory processes. Updates will be made available whenever significant updates to the sample database or analytical data become available.

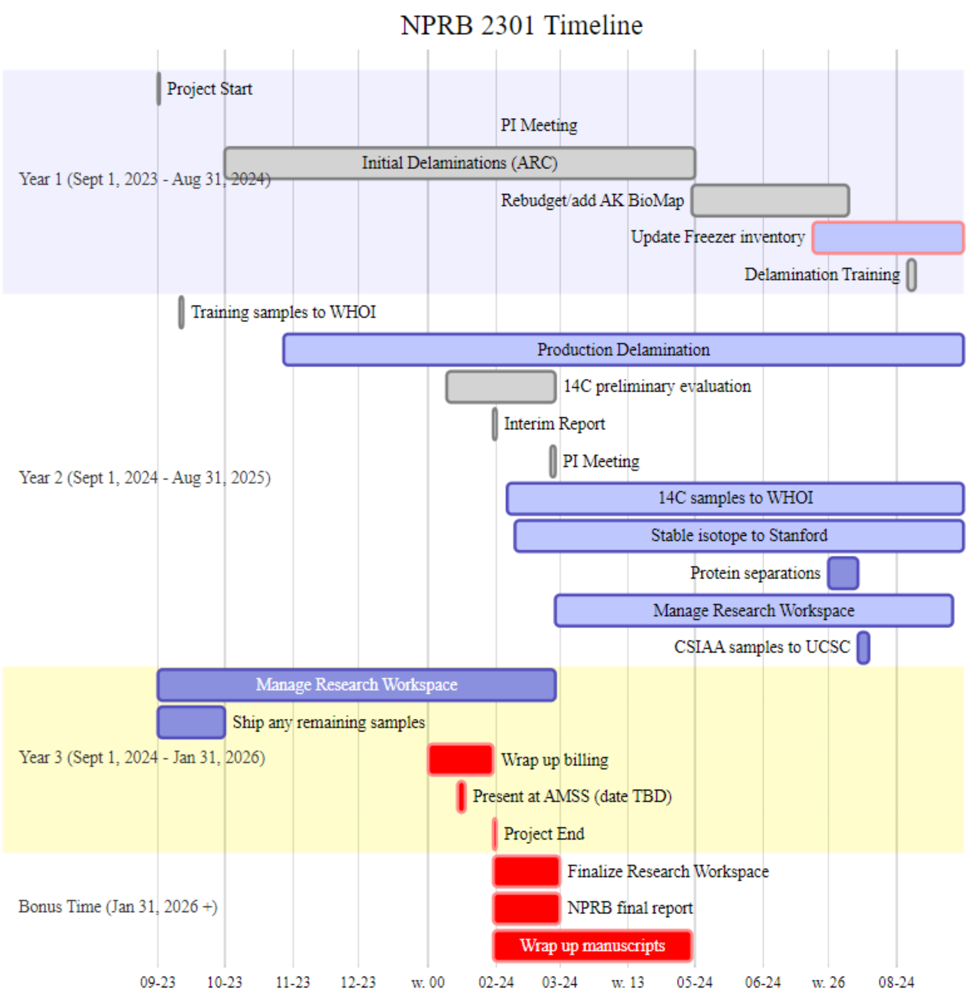
Version Notes

- 14C results updated (NPRB2301_layer_results), NOTE: this batch includes some samples from the freeze dried layers for comparison to the current method.
- Spiny dogfish spine reading in progress
- CSIA-AA samples in shipment
- Added more haul data, which changes some specimen ranks, fixed some minor data entry errors, see NPRB2301_specimen_summary
- Updated layer diameters for some layers

Links and Resources

- github repo (public)
- Research Workspace (password protected)
- Google Folder (access controlled)

Project Timeline



Specimen, Sample and Layer Summaries

- Each animal has a specimen_ID. The exception is spiny dogfish embryos, which are considered samples taken of the adult.
- Each eye, embryo, spine, etc. has a unique sample_id, which links back to the specimen_id that that sample was taken from.
- Each eye is further separated into layers and each layer_id links back to the sample_id (and thus, specimen_id).Layers are numbered sequentially from the outside towards the center. Early numbering convention used "N" for nucleus, and "R" for the remainder.
- Embryo each have a sample_id, but two eyes. Each eye and/or layer is labeled with the sample_id, L or R and layer number (as applicable).

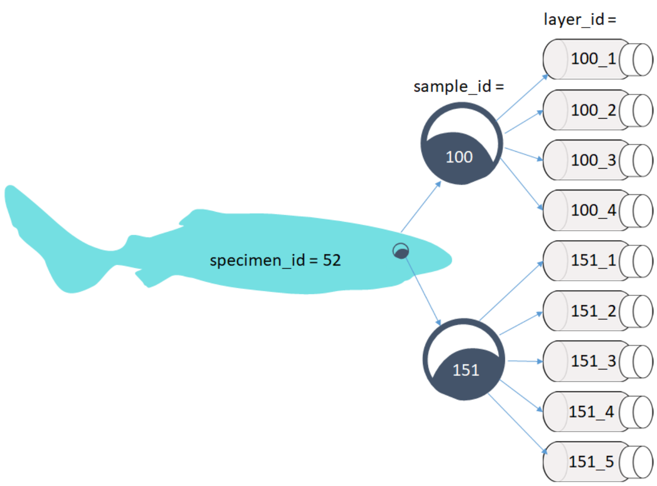


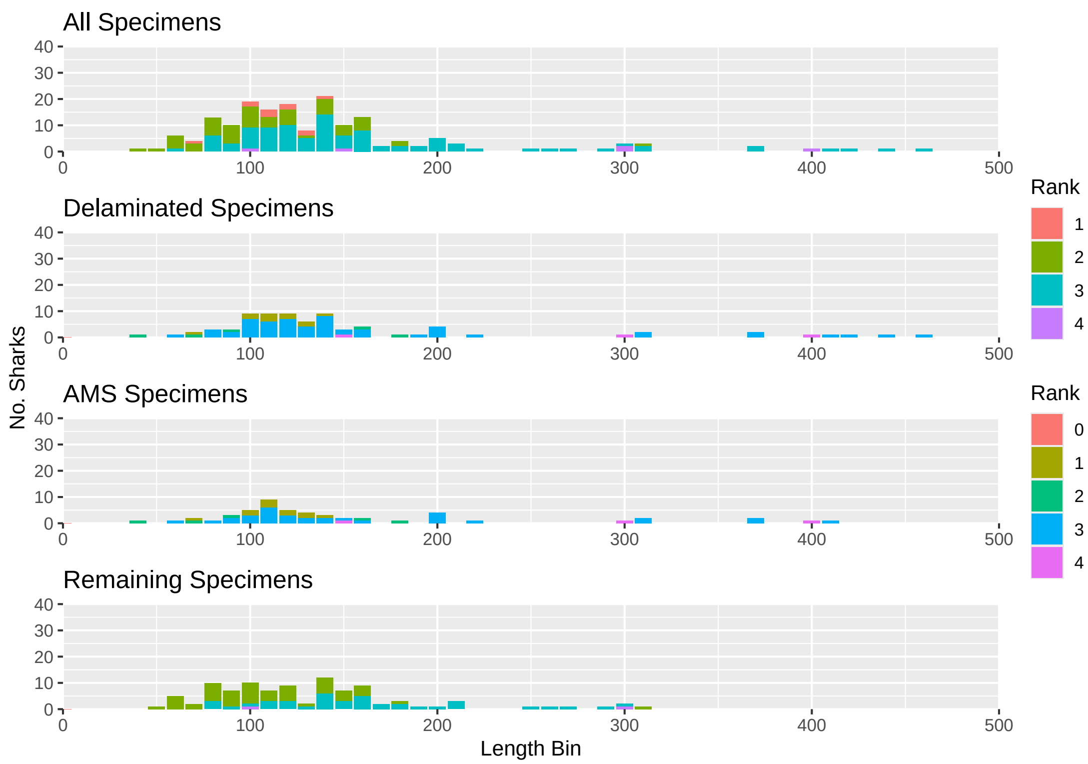
Figure 1: Specimens to samples to layers flow chart

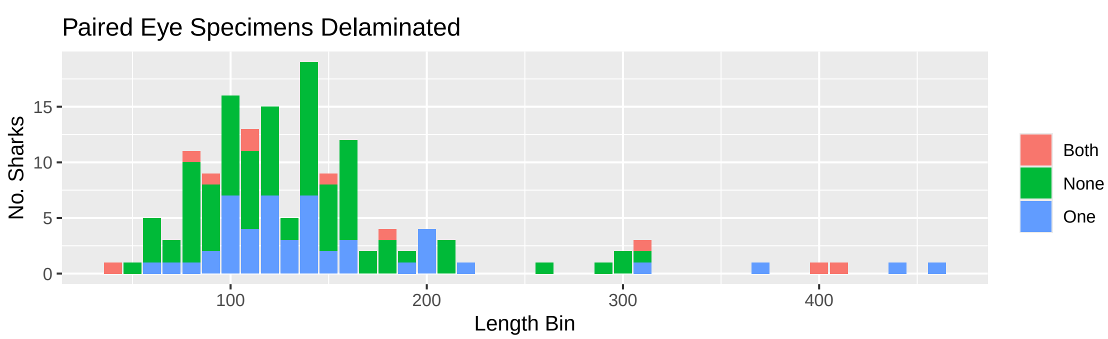
Pacific sleeper sharks

The below table only includes specimens with both length and haul data. There are some samples without haul data, and some without length. As data gaps get filled in, these numbers will change. Please request any other data breakdowns for future reports.

N PSS	N PSS eyes	N PSS Delaminated	N PSS Layers
172	319	98	517

For the purposes of the below graphs, all unknown length types are treated as Total Length until we resolve how to determine that. Ranks are from 1 (lowest) to 4 (highest) and described here. Rank = 1 are samples that had poor handling and may have thawed before delamination, Rank = 2 are collections with missing length or haul data, but do have at least a "source" (being who provided the sample), Rank = 3 are collections with complete data, but some uncertainties (i.e., length type), and Rank = 4 are survey collections with best chain of custody and full data available. Many specimens have not yet been attributed to a "source" in the database, this is intentional. Many of the at-sea observer provided samples did not have full data with the sample, or deck forms submitted. Some of these data gaps will be filled as we dig through at-sea observer databases, and those specimens will change from rank = 0 to the appropriate rank 1 - rank 4 value.





Spiny dogfish

N SD	N SD with full data	N SD available	N SD with samples but no length
22	11	5	5

There are two concerns with spiny dogfish samples:

1. Four spiny dogfish were processed before we identified potential contaminations and considered not available, however are included in the total number of spiny dogfish.
2. A subset of spiny dogfish do not have length or specific haul data (8). All of these samples do have both eyes and embryos and may still be useful for analyses between mother/offspring.

Began "dry peels" of spiny dogfish eye lens. These lens behave quite differently from the PSS lens. They are more solid, with clearly defined "petals" that peel away. Wil has been helping with these and confirms that the spiny dogfish eyes are much more similar to all of the teleost eyes that he has worked with. Working hypothesis is that the more visual predators may have more structural eyes.

Carbon 14

N 14C sent	N 14C returned	N 14C samples left per budget	Total N budgeted
271	241	319	560

14C results can be found in NPRB2301_layer_results.

Date Sent	N layers sent	N layers returned
Mar 25, 2019	11	11
Aug 5, 2019	8	8
Sept 11, 2024	36	35
Feb 6, 2025	70	70
Mar 31, 2025	96	95
Apr 28, 2025	50	21

Stable Isotopes

N SIA sent	N SIA returned	N SIA samples left per budget	Total N Budgeted
227	192	368	560

Date Sent	N layers sent	N layers returned
Feb 4, 2025	70	70
Mar 31, 2025	107	107
Apr 1, 2025	15	15
Apr 25, 2025	35	0

Stable isotope results can be found in NPRB2301_layer_results.

CSIA-AA

NOTE: we have funds for more samples than initially planned. Total sample size is now 30. We will proceed with the plan for the first 20, then discuss how to use the remaining samples. Plan: sequence layers (n=5) from either 507 or 508 (which ever has least layers for AMS) sequence layers (n=5) from 678 core and outer layer (n=2) from 161, 273, 422, 473, and 544

N CSIA_AA sent	N CSIA_AA returned	N CSIA_AA samples left per budget	Total N Budgeted
0	0	30	30